

Discrete Mathematics — Tutorial Sheet 05 — Relations

BSc (H) in Comp Sci, BSc (H) in Comp Foren

Intro/Onto and One-to-One

See questions in notes.

Question 1

For each of the following relations R defined on set $A = \{1, 2, 3, \dots\}$, determine which of the given ordered pairs belong to R

- (a) $(x, y) \in R$ iff $x|y$; $(2, 3), (2, 4), (2, 8), (2, 17)$
 (b) $(x, y) \in R$ iff $x \leq y$; $(2, 3), (3, 2), (2, 4), (5, 8)$
 (c) $(x, y) \in R$ iff $y = x^2$; $(1, 1), (2, 3), (2, 4), (2, 6)$

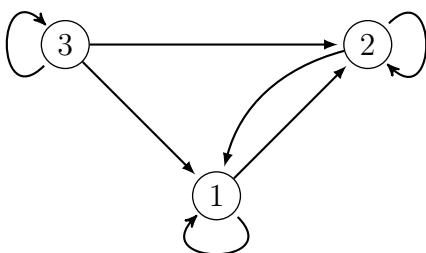
Properties of Relation on a Set

Question 2

Consider the relations represented in the following graphs.

- Determine whether the given relations are reflexive, symmetric, antisymmetric, or transitive.
- Determine which relations are asymmetric, irreflexive.
- Which of the graphs are of equivalence relations?
- Construct the transitive closure of each relation.

(a)



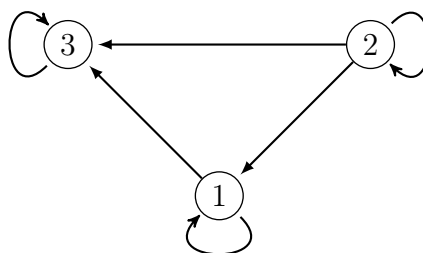
(d)

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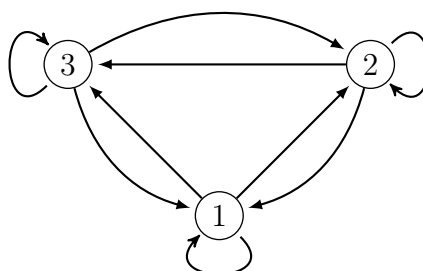
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1

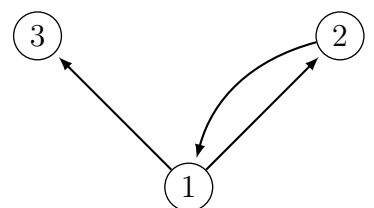
(b)



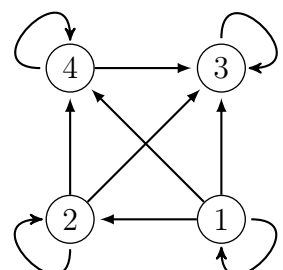
(e)



(c)



(f)



Question 3

Consider the relation on $\{1, 2, 3, 4, 5, 6\}$ defined by $R = \{(i, j) : |i - j| = 2\}$.

- (a) Is R reflexive?
- (b) Is R symmetric?
- (c) Is R transitive?
- (d) Draw a digraph of R .

Question 4

Determine which of the following are equivalence relations for the given sets:

- (a) $A = \{\text{lines in the plane}\}$, and R defined by $(x, y) \in R$ if and only if x is parallel to y .
- (b) $A = \mathbb{R}$ and relation R defined by $(x, y) \in R$ if and only if $|x - y| \leq 7$.